Cloud Computing: Resilience & DevOps

Module 8: Resilience, Observability & DevOps

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Outline

1. High Availability & Resilience
2. Backup & Disaster Recovery
3. Monitoring & Observability
4. DevOps & Infrastructure as Code





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Availability vs. Reliability

Availability

- System is up and running
- Measured as uptime percentage
- $99.9\% = 8.7$ hours down/year
- $99.99\% = 52$ min down/year

Reliability

- System works correctly
- Does it do what's expected?
- Can be available but unreliable
- Requires testing, validation

Both Matter

You need both high availability AND reliability for production systems.





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Achieving High Availability

Key Strategies

Redundancy at every layer: compute, storage, network, database.

Techniques:

- Multi-zone deployment: instances in different zones
- Load balancing: distribute traffic across instances

Goal

Design so that no single component failure brings down the whole system.



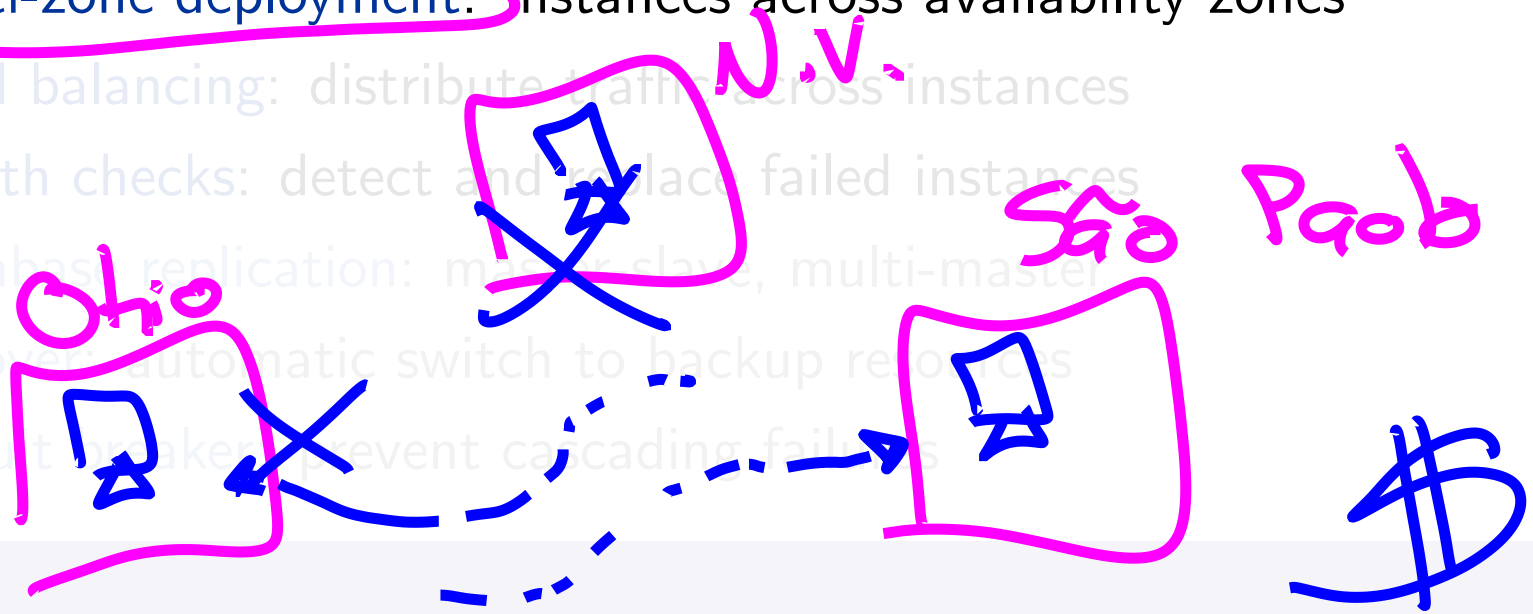
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- Multi-zone deployment: instances across availability zones
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- Database replication: master-slave, multi-master
- Failover: automatic switch to backup resources
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RPO and RTO

RPO: Recovery Point Objective

- How much data can you lose?
- Last backup was 1 hour ago
- `RPO` = 1 hour
- Impacts business loss

RTO: Recovery Time Objective

- How fast must you recover?
- Takes 4 hours to restore
- `RTO` = 4 hours
- Impacts downtime cost

Define Before Disaster

Know your `RPO`/`RTO` *before* something breaks. Test recovery regularly.





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DR Strategy Types

Cold Standby

- No running backup
- Data backed up offline
- Slower recovery
- Lower cost
- Hours to days

Warm Standby

- Partial backup running
- Replicated data in sync
- Faster than cold
- Medium cost
- Minutes to hours

Hot/Active-Active

- Fully running backup
- Real-time replication
- Instant failover
- Highest cost
- Seconds or less





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Disaster Recovery Strategies

Increasing cost and complexity →

Cold Standby	Warm Standby	Hot/Active-Active
Hours/days	*Minutes*	*Seconds*

Choice depends on business impact of downtime.





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The Three Pillars

Metrics

Numeric measurements: CPU, memory, requests/sec, latency, error rate.
Time-series data.

Logs

Textual events: application messages, errors, warnings.
Searchable, indexed.

Traces

Request flow: which services handled which operations, where time was spent.

Use All Three

Metrics show *what*, logs show *why*, traces show *how*.





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Service Level Indicator
Specific measurement (e.g., 99.5% uptime)

SLO
Service Level Objective
Target for SLI (e.g., "maintain 99.9%")

SLA
Service Level Agreement
Legal contract with penalties

Relationship

SLA is a contract. **SLO** is your internal target. **SLI** is what you measure.





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DevOps Philosophy

Breaking Silos

Development and Operations work together, not against each other.
Shared responsibility for deployment, reliability, and performance.

Key Practices:

- Continuous Integration: code merged frequently, tested automatically
- Continuous Deployment: passing tests auto-deploy to production





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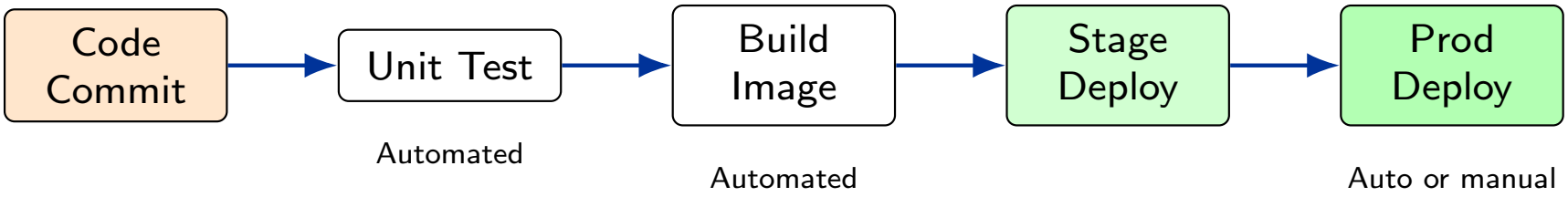
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CI/CD Pipeline



Infrastructure as Code (IaC)

Principle

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Benefits:

- Reproducibility: rebuild infrastructure identically
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Recap: Module 8

- **HA & Resilience**: redundancy at all layers, health checks, failover
- **Backup & DR**: define RPO/RTO, test recovery regularly
- **Monitoring**: metrics, logs, traces—all three matter
- **SLI/SLO/SLA**: measure performance, set targets, honor contracts
- **DevOps**: break silos between dev and ops
- **CI/CD**: automate testing and deployment
- **IaC**: version infrastructure like code





Thanks!

Questions?



Repo: *https://github.com/EngAndres/ud-public/tree/main/courses/cloud-computing*





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